

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2023

Subject SA2 – Life Insurance

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You are working in the Actuarial department of a medium sized Indian life insurance company writing unit-linked, non-participating savings and non-participating protection lines of business. The regulator is planning to abandon the existing '**Rules Based**' regulations and instead adopt a '**Principles Based**' regulations covering the "Product Development and Pricing" and the "Solvency Assessment" areas.

As a first step, the regulator has requested insurance companies to draft a Board approved framework, capturing the key policies and procedures to be followed by the managements of companies covering the "Product Development and Pricing" of life insurance contracts.

- i)** Describe the key policies and procedures that should be captured in such a Board approved policy. (18)

Further, the regulator is planning to replace the existing '**Factors Based**' solvency assessment approach to that based on a '**Risk Based Capital (RBC)**' requirement. In this regard, the regulator has released the guidelines for insurance companies to perform several "Quantitative Impact Studies (QISs)", wherein the insurance companies are required to calibrate their own stress factors for all the economic and non-economic risks.

- ii)** Identify, with reasons, the key risks for each line of business of your Company that could have significant contribution towards the new RBC requirement and describe how such risks would have been allowed for in the assessment of solvency in the current Factors Based regime. (17)

Based on the QISs, the solvency ratio of the Company has improved to 250%, from 200% (basis the existing solvency regime). Further, the sensitivity of the solvency ratio to changes in interest rate (say 100bps parallel movement in existing yield curve) has increased from 2% points (as per the current solvency regime) to 15% points in the proposed RBC regime.

- iii)** Discuss the reasons for the increase in solvency ratio and the increased sensitivity of the solvency ratio when moving from the current solvency regime to the proposed RBC approach. (15)
[50]

- Q. 2)** ABC Life Insurance Company is planning to introduce the following two products:

- (a) A participating deferred annuity product, with the annuity rates (separate for standard and impaired lives) being guaranteed for life, provided at the inception of the deferred annuity contract; and
- (b) A non-participating protection product wherein the benefit is payable in the form of an annuity for life, upon either the diagnosis of any of the defined critical illnesses; or inability to perform any three of the six defined Activities of Daily Living (ADLs), whichever happens first.

During the training, the Company's agents asked to explain the difference between the two products as both the products provide annuity payments. They also wanted to understand how the underwriting requirements for these two products would differ from each other.

- i)** Discuss how the two products cater to different needs of the customers. (10)
- ii)** Discuss the key risks faced by the Company in these two products and how the underwriting requirements would differ between the two products. (20)

The Sales Manager has enquired why the surrender values proposed in the participating deferred annuity product is low despite having lower overall commission rates in this product as compared to other participating endowments. She has suggested to increase the surrender values, so that the product becomes competitive.

- iii)** Explain the possible reasons for low surrender values and how the Company can offer competitive surrender values in this product.

(8)

The Company did not increase the proposed surrender values as suggested by the Sales Manager. After two years of introduction of the product, the Company started to experience a decline in the new business volumes and also high surrenders in the participating deferred annuity product. Upon further analysis, it was observed that this was due to an increase in the interest rates for periods up to 10 years, making the other financial products in the country more attractive for the customers.

- iv)** Discuss the possible steps that the Company may take to make the participating deferred annuity product more attractive to the customers.

(12)

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